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Unit 3 Repetition Statements "While / Until"

PDF Documents

PDF Document 1: 1731842097319-Chapter 3_ unit3 Repetition Statements.pdf

This PDF document is included in the unit materials.

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UNIT 3 REPETITION STATEMENTS

While statement

A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

✓ WHILE



condition
While the sun is up,



task
Keep the LED off

Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing."



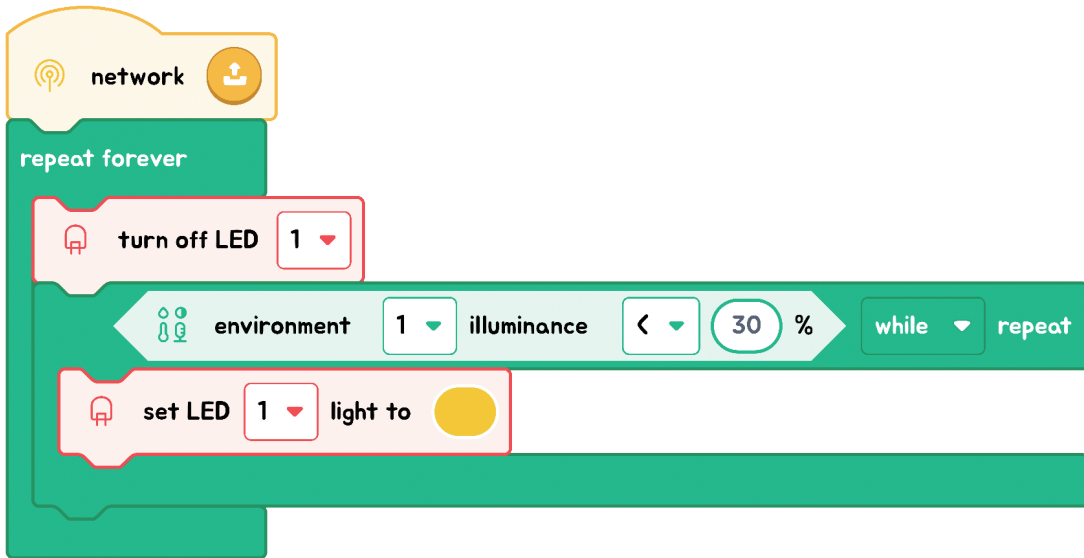
condition
While the timer is running,



task
Keep brushing your teeth.



EXPLANATION WITH CODE EXAMPLE



In our code example, we're working with a light sensor and an LED. When we start, our LED light is turned off. But we have a special rule written in our code about when to turn it on and make it yellow.

1) Is the illuminance level below 30%?

This checks how bright the light around us is. If it's really dim, less than 30% bright, our rule is met.

2) Set the LED to yellow.

Once we know the light is dim enough, our code tells the LED to turn on and shine in a yellow color. This makes the LED light up yellow as long as the room stays dim.



Code Insight

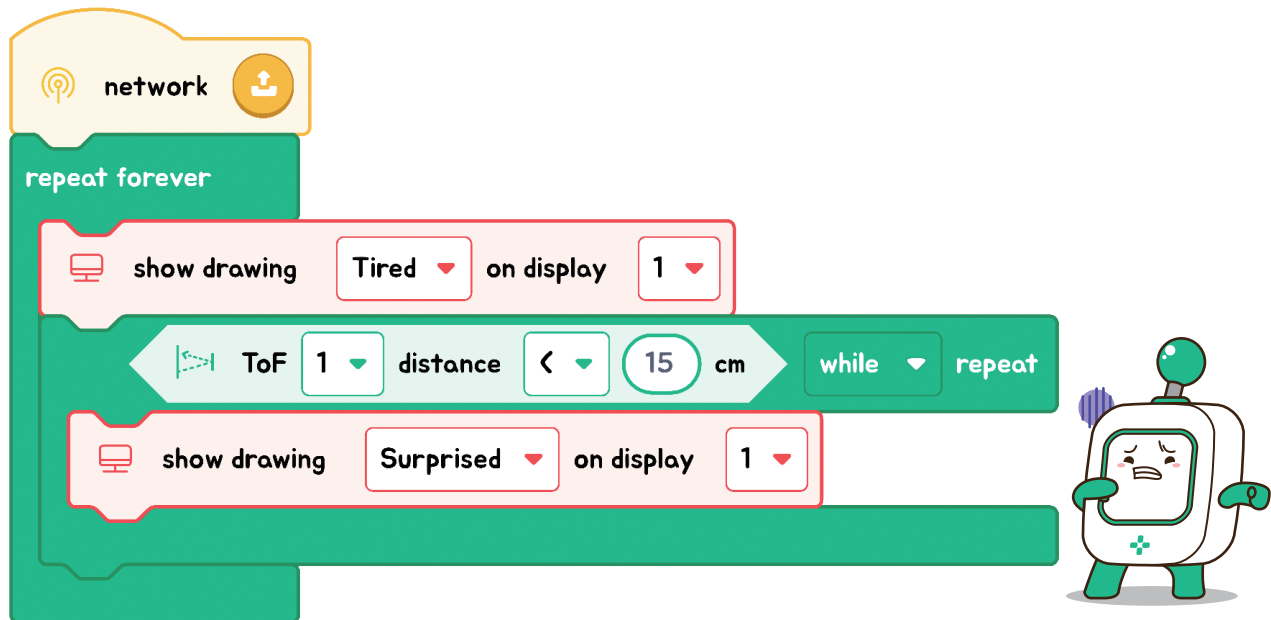
- 1. Default State:** Initially, the LED is off. This state persists **until the "while" loop starts and when the illuminance condition isn't met.**
- 2. Condition Checking:** The code first checks **if the light level is below 30%**, determining the next steps.
- 3. Execution While True:** With light levels under 30%, the LED turns yellow, repeating this task as long as the dim condition lasts.
- 4. Exiting the Loop:** **When light levels exceed 30%, the loop ends**, and the LED shuts off, awaiting the next dim condition.



Exercise

Exercise 1 : Surprise MODI !

Imagine we're trying to surprise a sleepy MODI robot. When we get close to MODI, it acts shocked. Try to spot the 'condition' that makes MODI react and the 'action' MODI takes in the code.



Exercise 2 : Knowledge Check

1 What's the main use of a while loop in coding?

- ☐ A To run a section of code just once if a certain check is true.
- ☒ B To keep running a section of code over and over as long as a certain check is true.
- ☐ C To run a section of code no matter what.
- ☐ D To run a section of code once and then stop no matter what.

2 When does a while loop finish its job?

- ☒ A When the check inside the loop turns out false.
- ☐ B When the check inside the loop turns out true.
- ☐ C If the loop runs into a problem.
- ☐ D When the loop gets to a certain point in the code.

Until statement

In programming, the "until" statement is like having a goal and doing something over and over until you reach that goal.

✓ **UNTIL**

\$1000



condition

Until you reach
your goal of \$1,000,

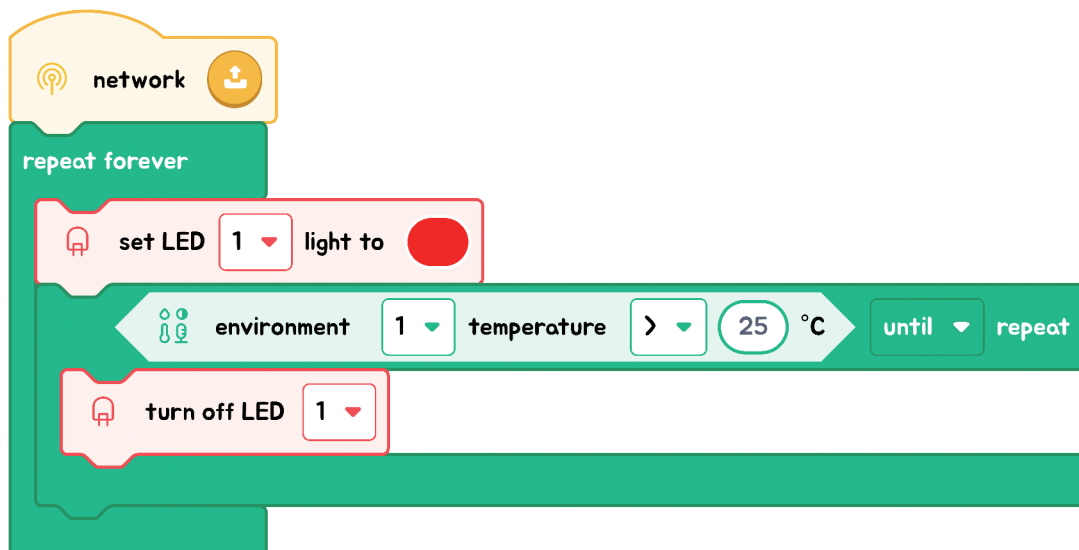
task 1

Keep saving money.

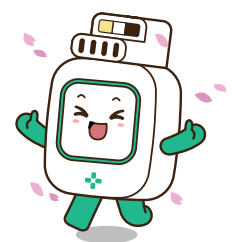
For example, if your goal is to save \$1,000, you keep adding money to your savings "until" you have \$1,000. Just like saving money bit by bit until you hit your target, the "until" statement repeats an action until a certain condition is met.

Another Example:

In our example, think of the LED as a light that stays off "until" it gets warm enough, say 25 degrees. So, the LED doesn't turn on while it's cooler.



But once it hits 25 degrees or more, the red light comes on. The "until" statement here waits for the temperature to go up, and when it does, it's like a signal to turn the red light on.





Exercise

Exercise 1 : Surprise MODI !

What is the primary difference between "until" and "while" loops in block coding?

- ☐ A "Until" loops execute the code block until a condition is true, while "while" loops execute the code block as long as a condition is true.
- ☐ B "Until" loops execute the code block as long as a condition is true, while "while" loops execute the code block until a condition is true.
- ☐ C "Until" loops and "while" loops both execute the code block until a condition is true.
- ☐ D There is no difference between "until" and "while" loops.

Exercise 2 : Knowledge Check

Change the while statement below to an "until" statement.

